

Square Roots (Vargmool)

Aasan Hinglish Notes, Methods, Examples aur Practice Questions

Square root, square (varg) ka ulta hota hai. Is guide me concept ko simple Hinglish me samjhaya gaya hai, methods diye gaye hain, fir 10 SOLVED EXAMPLES (sabse HARD se sabse EASY tak, har level cover karte hue), aur end me 22 practice questions ke STEP-BY-STEP solutions diye gaye hain.

EXAM * aise green tag wale topics exams me baar-baar aate hain

1. Square Root kya hota hai?

Definition Kisi number ka square root wo value hai jise khud se multiply karne par wahi number mil jaye. Ye squaring ka ulta (inverse) hai.

Symbol: Radical sign $\sqrt{\quad}$ use hota hai. Jaise $\sqrt{25}$.

Examples:

- $\sqrt{25} = 5$ kyunki $5 \times 5 = 25$
- $\sqrt{81} = 9$ kyunki $9 \times 9 = 81$
- Square: $6 \times 6 = 36$ $\rightarrow$ Square root: $\sqrt{36} = 6$

2. Perfect Squares

EXAM * Exam me CONSTANTLY aata hai

Definition Perfect square wo number hota hai jiska square root ek pura (whole) number ho.

Examples: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144 ...

Yaad rakho:

- Perfect squares ke square root exact whole numbers hote hain.
- 2, 3, 5, 7 jaise numbers perfect square nahi hain; inke root non-ending decimals (irrational) hote hain, jaise $\sqrt{2} = 1.414...$
- Pehchaan: perfect square ke end me sirf 0, 1, 4, 5, 6 ya 9 aata hai.
- Perfect square kabhi 2, 3, 7 ya 8 par khatam NAHI hota.

Squares Table (1 se 20) - yaad karo:

$1^2=1$	$2^2=4$	$3^2=9$	$4^2=16$	$5^2=25$
$6^2=36$	$7^2=49$	$8^2=64$	$9^2=81$	$10^2=100$
$11^2=121$	$12^2=144$	$13^2=169$	$14^2=196$	$15^2=225$
$16^2=256$	$17^2=289$	$18^2=324$	$19^2=361$	$20^2=400$

3. Square Root Nikalne ke Methods

EXAM * Exam me CONSTANTLY aata hai

(a) Prime Factorisation Number ko prime factors me todo, pairs banao, aur har pair me se ek factor lo.

- Example: $144 = 2 \times 2 \times 2 \times 2 \times 3 \times 3 \rightarrow 2 \times 2 \times 3 = 12$, to $\sqrt{144} = 12$

(b) Long Division Bade numbers aur non-perfect squares ke liye step-by-step answer (decimals me bhi) nikalne ke liye.

(c) Estimation Do nazdeeki perfect squares dekh kar approximate value guess karo.

- Example: $\sqrt{150}$, $\sqrt{144}=12$ aur $\sqrt{169}=13$ ke beech hai, to lagbhag 12.2

4. Kaam ki Properties

- $\sqrt{a \times b} = \sqrt{a} \times \sqrt{b}$
- $\sqrt{a / b} = \sqrt{a} / \sqrt{b}$
- $\sqrt{0} = 0$ aur $\sqrt{1} = 1$
- $(\sqrt{a})^2 = a$ aur $\sqrt{a^2} = a$
- Negative number ka koi real square root nahi hota.

4B. Pythagorean Triplets

EXAM * Exam me CONSTANTLY aata hai

Kya hai: Teen numbers (a, b, c) jinme $a^2 + b^2 = c^2$ ho. Right-angle triangle me kaam aate hain.

- Famous triplets: (3, 4, 5), (5, 12, 13), (8, 15, 17), (7, 24, 25).
- Check: $3^2 + 4^2 = 9 + 16 = 25 = 5^2$. Sahi!
- Kisi triplet ko same number se multiply karne par naya triplet milta hai: (3, 4, 5) $\times 2 =$ (6, 8, 10).

5. 100 se 200 ke Beech Perfect Squares

Is range me sirf paanch perfect squares hain:

$\sqrt{100} = 10$	$(10 \times 10 = 100)$
$\sqrt{121} = 11$	$(11 \times 11 = 121)$
$\sqrt{144} = 12$	$(12 \times 12 = 144)$
$\sqrt{169} = 13$	$(13 \times 13 = 169)$
$\sqrt{196} = 14$	$(14 \times 14 = 196)$

6. Solved Examples - Hard se Easy (Sab Levels)

Neeche 10 examples step-by-step solve karke dikhaye gaye hain - sabse MUSHKIL se shuru karke sabse AASAN tak, taaki har type ka method clear ho jaye.

Example 1 (Hardest): Long Division se $\sqrt{5476}$

- Step: Right se do-do digit ke pair banao: 54 | 76.
- Step: Pehla pair 54: sabse bada n jiska $n^2 \leq 54 \rightarrow 7$ (49). Remainder = 5.
- Step: Next pair niche lao: 5 76 $\rightarrow$ 576.
- Step: Quotient (7) ko double karo = 14. '14_' $\times$ _ $\leq$ 576 dhoondo.
- Step: $144 \times 4 = 576$ (exact!). Dusra digit = 4.
- Step: Quotient = 74.

Answer: $\sqrt{5476} = 74$

Example 2 (Hard): Decimal ka root - $\sqrt{0.0144}$

- Step: 0.0144 ko fraction banao: $144 / 10000$.
- Step: $\sqrt{144} = 12$, $\sqrt{10000} = 100$.
- Step: $= 12 / 100$.

Answer: $\sqrt{0.0144} = 0.12$

Example 3 (Hard): Surd simplify - $\sqrt{72}$

- Step: 72 ko perfect square x baaki me todo: $72 = 36 \times 2$.
- Step: $\sqrt{72} = \sqrt{36} \times \sqrt{2} = 6 \times \sqrt{2}$.

Answer: $6\sqrt{2}$ (lagbhag 8.49)

Example 4 (Medium-hard): Prime factorisation - $\sqrt{1296}$

- Step: $1296 = 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 3 \times 3 = 2^4 \times 3^4$.
- Step: Pairs banao: $(2 \times 2)(2 \times 2)$ aur $(3 \times 3)(3 \times 3)$.
- Step: Har pair me se ek lo: $2^2 \times 3^2 = 4 \times 9$.

Answer: $\sqrt{1296} = 36$

Example 5 (Medium-hard): Property - $\sqrt{144 \times 25}$

- Step: Rule: $\sqrt{a \times b} = \sqrt{a} \times \sqrt{b}$.
- Step: $= \sqrt{144} \times \sqrt{25} = 12 \times 5$.

Answer: $\sqrt{144 \times 25} = 60$

Example 6 (Medium): Prime factorisation - $\sqrt{324}$

- Step: $324 = 4 \times 81 = 2^2 \times 3^4$.
- Step: Har pair me se ek lo: $2 \times 3^2 = 2 \times 9$.

Answer: $\sqrt{324} = 18$

Example 7 (Medium): Estimation - $\sqrt{50}$ kin ke beech?

- Step: Nazdeeki perfect squares: $7^2 = 49$ aur $8^2 = 64$.
- Step: 50, 49 aur 64 ke beech hai -> answer 7 aur 8 ke beech.

Answer: Lagbhag 7.07 (7 aur 8 ke beech)

Example 8 (Easy-medium): Fraction - $\sqrt{49/64}$

- Step: Rule: $\sqrt{a/b} = \sqrt{a} / \sqrt{b}$.
- Step: $= \sqrt{49} / \sqrt{64} = 7 / 8$.

Answer: $\sqrt{49/64} = 7/8$

Example 9 (Easy): Perfect square - $\sqrt{169}$

- Step: Koi number dhoondo jiska square 169 ho.
- Step: $13 \times 13 = 169$.

Answer: $\sqrt{169} = 13$

Example 10 (Easiest): $\sqrt{0}$ aur $\sqrt{1}$

- Step: $0 \times 0 = 0$ -> $\sqrt{0} = 0$.
- Step: $1 \times 1 = 1$ -> $\sqrt{1} = 1$.

Answer: $\sqrt{0} = 0$, $\sqrt{1} = 1$

7. Practice Questions

Pehle khud solve karne ki koshish karo. Neeche har question ka step-by-step solution diya gaya hai.

Q1. $\sqrt{49} = ?$

Q2. $\sqrt{64} = ?$

Q3. $\sqrt{81} = ?$

Q4. $\sqrt{100} = ?$

Q5. $\sqrt{121} = ?$

Q6. $\sqrt{144} = ?$

Q7. $\sqrt{169} = ?$

- Q8. $\sqrt{196} = ?$
Q9. $\sqrt{225} = ?$
Q10. $\sqrt{256} = ?$
Q11. Kya 150 ek perfect square hai?
Q12. $\sqrt{180}$ kin do whole numbers ke beech aata hai?
Q13. $\sqrt{36} + \sqrt{64} = ?$
Q14. Ek square park ka area 196 sq m hai. Uski side kitni hogi?
Q15. 100 se 200 ke beech kaunse perfect squares hain?
Q16. $\sqrt{2}$ rational hai ya irrational?
Q17. $\sqrt{400} = ?$
Q18. $\sqrt{1} = ?$ aur $\sqrt{0} = ?$
Q19. Kya negative number ka real square root hota hai?
Q20. $\sqrt{9 \times 16} = ?$ (property use karo)
Q21. $\sqrt{625} = ?$
Q22. $12 \times 12 = 144$, to $\sqrt{144} = ?$

8. Step-by-Step Solutions

Q1. $\sqrt{49}$

- Step: Koi number dhoondo jiska square 49 ho.
- Step: $7 \times 7 = 49$.

Answer:7

Q2. $\sqrt{64}$

- Step: $8 \times 8 = 64$.

Answer:8

Q3. $\sqrt{81}$

- Step: $9 \times 9 = 81$.

Answer:9

Q4. $\sqrt{100}$

- Step: $10 \times 10 = 100$.

Answer:10

Q5. $\sqrt{121}$

- Step: $11 \times 11 = 121$.

Answer:11

Q6. $\sqrt{144}$

- Step: $12 \times 12 = 144$.

Answer:12

Q7. $\sqrt{169}$

- Step: $13 \times 13 = 169$.

Answer:13

Q8. $\sqrt{196}$

- Step: $14 \times 14 = 196$.

Answer:14

Q9. $\sqrt{225}$

- Step: $15 \times 15 = 225$.

Answer:15

Q10. $\sqrt{256}$

- Step: $16 \times 16 = 256$.

Answer:16

Q11. Kya 150 perfect square hai?

- Step: $12^2 = 144$ aur $13^2 = 169$.
- Step: 150 in dono ke beech hai, koi whole root nahi.
- Step: $\sqrt{150} \sim 12.25$.

Answer:Nahi

Q12. $\sqrt{180}$ kin do whole numbers ke beech?

- Step: $13^2 = 169$ aur $14^2 = 196$.
- Step: 180 in dono ke beech aata hai.

Answer:13 aur 14 ke beech

Q13. $\sqrt{36} + \sqrt{64}$

- Step: $\sqrt{36} = 6$, $\sqrt{64} = 8$.
- Step: $6 + 8$.

Answer:14

Q14. Square park ka area 196 sq m, side?

- Step: Square ka area = side x side, to side = $\sqrt{\text{area}}$.
- Step: side = $\sqrt{196}$.

Answer:14 metre

Q15. 100 se 200 ke beech perfect squares?

- Step: $10^2=100$, $11^2=121$, $12^2=144$, $13^2=169$, $14^2=196$.

Answer:100, 121, 144, 169, 196

Q16. $\sqrt{2}$ rational ya irrational?

- Step: $\sqrt{2} = 1.41421...$ non-terminating aur non-repeating hai.

Answer:Irrational

Q17. $\sqrt{400}$

- Step: $20 \times 20 = 400$.

Answer:20

Q18. $\sqrt{1}$ aur $\sqrt{0}$

- Step: $1 \times 1 = 1$ aur $0 \times 0 = 0$.

Answer: $\sqrt{1} = 1$, $\sqrt{0} = 0$

Q19. Negative number ka real square root?

- Step: Kisi bhi real number ka square negative nahi hota.
- Step: Isliye negative ka real square root possible nahi.

Answer:Nahi hota

Q20. $\sqrt{9 \times 16}$ (property)

- Step: Rule: $\sqrt{a \times b} = \sqrt{a} \times \sqrt{b}$.
- Step: $= \sqrt{9} \times \sqrt{16} = 3 \times 4$.

Answer:12

Q21. $\sqrt{625}$

- Step: $25 \times 25 = 625$.

Answer:25

Q22. $12 \times 12 = 144$, to $\sqrt{144}$?

- Step: sqrt square ka ulta hai; 12 ka square 144 hai.

Answer:12